Leo Zhang

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Professional Summary

Master's in Computer and Information Sciences from AUT, **First Class Honours**. **Integration** Developer — Backend engineer with **10+ years** building enterprise systems across **Java**, Spring Boot and cloud infrastructure. Focused on software architecture, API design and building services that balance **quality** with **delivery** constraints. Delivered end-to-end features across web, mobile and IoT with hands-on CI/CD and **automated testing**. Published AI research at IVCNZ 2025, with practical experience integrating LLM workflows into **delivery** pipelines. **Delivery** focus includes observability and stable release **practices**.

Core Competencies

Backend Engineering: Java, Spring Boot, Spring Cloud, RESTful **APIs**

API & System Design: API Design (REST/GraphQL), System Architecture, Event-Driven Architecture, Distributed Systems

Cloud & Infrastructure: AWS (ECS, Lambda, S3), Docker, Kubernetes, CI/CD

Databases & Storage: MySQL, PostgreSQL, MongoDB, Redis

Languages & Tools: Python, TypeScript, C#/.NET, Cursor

Experience

Auckland University of Technology — *Auckland, New Zealand* 2024 – 2026

AI Research Engineer

- Designed and delivered ChatClothes, a digital multimodal AI platform integrating computer vision, diffusion models, and LLM-powered workflows with RAG, vector search, and prompt grounding. Conducted extensive image dataset work, model training and **testing** across multiple CV projects. Provided technical leadership and owned the product end-to-end as cloud-native AI **solutions**, from system design and development through evaluation, deployment, and user validation.
- Deployed and optimized AI models on edge devices including Raspberry Pi and Android phones, running lightweight LLMs and vision models on resource-constrained ARM hardware with offline inference, image processing, and on-device prediction. Validated model outputs through ground truth **testing** and evaluation.

Tech: Python, PyTorch, Ollama, Docker, Linux, AWS, Azure, Git, Raspberry Pi, Android

Chunxiao Technology Co., Ltd. — *China* 2013 – 2024

*Technology company (est. 2013, ~40 employees) delivering software, hardware, and system **integration** across streaming media, enterprise platforms, IoT, and industrial automation.*

> **Senior Software Engineer | 2018 - 2024**
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> - Smart Factory IoT Platform: architected manufacturing platform across 10+ factory sites with 6-person cross-functional team. Integrated RFID readers, barcode scanners, electronic scales (RS232/RS485 serial bridge to WebSocket), conveyors, and Android workshop terminals with Spring Cloud microservices. Event-driven architecture via ActiveMQ/Kafka, Docker deployment, 30%+ efficiency gain.
> - IoT Device Management Platform: built Android device discovery/binding/remote control app and gateway firmware for smart switches and Zigbee/WiFi gateways. Protocol handling, MQTT cloud uplink, Spring Cloud backend API, and Vue.js management dashboard.

- Boobit Crypto Trading App: sole Android developer for financial cryptocurrency trading app. Real-time market data via WebSocket, order book rendering, candlestick charting, encrypted local wallet storage, MVVM with Clean Architecture, Retrofit/OkHttp, Room — delivering banking-grade reliability for sensitive financial transactions.
- Picture Book Locker: Android cabinet control system with electromagnetic locks, UV sterilization, sensors, face recognition + QR code authentication. Borrowing client app with <3s checkout time, 24/7 self-service.

Tech: **Kotlin, Java, Android SDK, Spring Cloud, REST APIs, MySQL, Redis, Docker, CI/CD, WebSocket, NDK/JNI, MQTT, Zigbee, RFID**

Senior Android Engineer | 2013 - 2018

- Enterprise Messaging Platform: architected Android client for 5,000+ DAU and 500K+ daily messages with sub-200ms latency. Built NDK/JNI TCP/UDP transport layer, WebSocket connection persistence, offline message sync, JPush backup push channel, and OEM ROM keep-alive strategies (Xiaomi/Huawei/OPPO). Migrated core IM from unstable in-house C++ to Easemob cloud, eliminating 90%+ defects. Built FastDFS distributed file storage (100GB+). Applied hotfix framework (Tinker, AndFix) for emergency patches and performance **quality** engineering across cold start, ANR, OOM, and APK size.
- Live Streaming Commerce System: full-stack Android + web **delivery** for live commerce platform. Android client in Java/Kotlin with RTMP/WebRTC/HLS streaming, multi-language architecture (C# + C++ media core + Python analytics + Lua config), 1000+ peak concurrency, 99.5% stream availability.

Tech: **Kotlin, Java, Android SDK, NDK/JNI, C++, WebSocket, TCP/UDP, Gradle, CI/CD, Tinker, AndFix, JPush, RTMP, WebRTC, HLS, Ea**

Education

Master of Computer and Information Sciences *Auckland University of Technology, NZ* 2024 – 2026

Research: Computer Vision, Diffusion Models, Large Language Models, Multimodal AI

Bachelor of Software Engineering *Hebei Univ. of Sci. & Tech., China* 2009 – 2013

Coursework: Data Structures, Algorithms, Java Programming, Database Systems, Software Engineering

Licenses & Certifications

IoT Fundamentals — Xi'an Jiaotong University 2020

Science and Technology Achievement Award (Smart Manufacturing) — Hebei, China 2020

Software Design Engineer Certificate — China Qualification Authority 2019

Interests

Hiking, self-hosted infrastructure, AI tools, and mobile technology trends

Referees

References available upon request.